Chemistry 8: Elements and Isotopes

CHEAT SHEET

@nova.xplains

What is an element?



Element = substance made of only one kind of atom

Atomic number = number of protons: Au 79, C 6, O 8

118 elements known, about 90 occur in nature

Chemical symbols

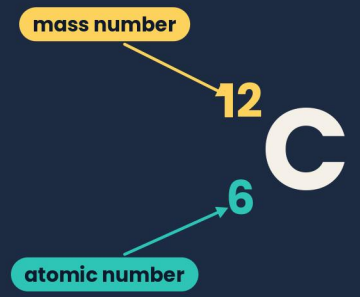


1 or 2 letters, the first one is a capital: H, He, Cl

H = hydrogen, C = carbon, O = oxygen

From Latin: Au (aurum), Fe (ferrum), Na (natrium), Ag (argentum)

Atomic & mass number



Mass number (top left) = protons + neutrons

Atomic number (bottom left) = protons

Neutrons = mass number - atomic number: 12 - 6 = 6

Isotopes



Same number of protons, different number of neutrons

Carbon: C-12 (6 n), C-13 (7 n), C-14 (8 n)

Chlorine 35.45 = mix of Cl-35 (76 %) and Cl-37 (24 %)

Heavy water

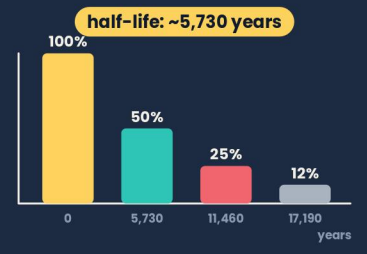


Hydrogen H: 1 p, 0 n · deuterium D: 1 p, 1 n · tritium T: 1 p, 2 n

About 1 in 6,400 hydrogen atoms is deuterium

D2O is about 11 % heavier: its ice sinks in normal water!

Radioactive isotopes



Unstable nuclei decay and send out radiation

C-14 half-life: about 5,730 years (radiocarbon dating)

Ötzi: about 5,300 years ago · your body: C-14, K-40

BONUS Quiz: 6 protons + 8 neutrons = carbon 14 (answer A). Next episode: reading the periodic table.

Sources: IUPAC (118 elements, standard atomic weights); NIST isotope data; radiocarbon dating (C-14 half-life 5,730 years); South Tyrol Museum of Archaeology (Ötzi)

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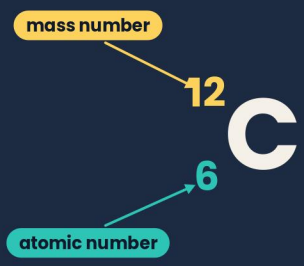


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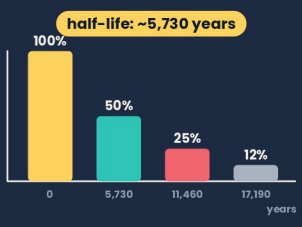


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